

Interaction Structure and Initiative Control in AI-Assisted Decision Support Systems

Anna Asatryan

Akian College of Science & Engineering
American University of Armenia
Yerevan, Armenia
Email: anna_asatryan2@edu.aua.am

Karen Hovhannisyan (Supervisor)

Akian College of Science & Engineering
American University of Armenia
Yerevan, Armenia
Email: khovhannisyan@aua.am

Abstract—This project investigates how the *timing of AI advice* affects decision quality in human–AI decision support systems (DSS). While predictive models are typically evaluated using offline metrics such as accuracy and calibration, real-world outcomes depend on how human decision-makers interact with model outputs.

We construct a controlled experimental DSS using a calibrated logistic regression model for credit default prediction and a model-independent notion of *case difficulty* derived from empirical base rates over discretized feature space. Holding the predictive model fixed, we experimentally manipulate the interaction protocol (AI-first, human-first, no-AI) and measure how these protocols affect reliance behavior and the translation of model-level performance into system-level outcomes.

We additionally introduce a normative decision benchmark based on asymmetric decision costs to evaluate alignment between human–AI decisions and cost-optimal policies, while distinguishing this benchmark from the model’s fixed decision threshold used for prediction.

Index Terms—Human-AI Interaction, Decision Support Systems, Interaction Timing, Reliance, Calibration, Decision Theory.

I. INTRODUCTION

Predictive models are typically evaluated using offline performance metrics such as accuracy, AUC, or calibration error. However, in deployed systems, predictions influence outcomes only through human decisions. As a result, identical predictive models can yield substantially different real-world outcomes depending on how recommendations are presented and incorporated into decision processes.

This project investigates how *interaction structure* mediates the translation of model-level performance into system-level outcomes. Interaction structure is treated as a controllable system variable governing both the timing and representation of AI advice.

In this work, interaction structure is defined primarily by the *timing of AI advice*. Specifically, we study whether AI predictions are presented before or after human judgment, and how this ordering affects reliance behavior and decision outcomes.

To enable controlled experimentation, we introduce a model-independent notion of *case difficulty*, defined using empirical base rates over discretized feature space. This avoids circularity between model predictions and evaluation, and

allows difficulty to be treated as an exogenous property of the decision environment.

The central objective is not to improve predictive accuracy, but to evaluate how effectively model predictions are translated into improved decisions under different interaction protocols.

The primary contribution of this work is the explicit treatment of interaction structure as a post-model evaluation layer, enabling systematic analysis of how human–AI interaction affects the realized performance of predictive systems independently of model accuracy.

II. LITERATURE REVIEW

[This section will summarize and analyze relevant scientific literature, including key existing approaches to human-AI interaction, reliance asymmetry, and decision support systems. It will address the research gap regarding how interaction timing and uncertainty representation impact system-level performance. Length: 1–4 pages.]

III. METHODOLOGY

A. Hypotheses

- **H1 (AI Benefit):** AI assistance improves final decision accuracy relative to unaided human decisions.
- **H2 (Difficulty Moderation):** The effect of AI assistance varies across difficulty levels, with different patterns of reliance in easy, medium, and hard cases.
- **H3 (Interaction Timing Effect):** The timing of AI advice (AI-first vs human-first) significantly affects reliance behavior and decision accuracy.
- **H4 (Selective Reliance):** Participants rely more on AI predictions when the model is correct than when it is incorrect.
- **H5 (Confidence Alignment):** Model confidence is imperfectly aligned with correctness, creating a non-degenerate environment in which reliance decisions are consequential. We expect interaction protocols that surface uncertainty explicitly to improve participant confidence calibration.
- **H6 (Normative Deviation):** Participants systematically deviate from cost-sensitive optimal decisions, particularly in medium-difficulty cases.

B. Data and Feature Engineering

The dataset consists of approximately 1.3 million LendingClub loans, with a binary default prediction task and an observed default rate of approximately 20%.

Only features available at loan origination are used to prevent data leakage. A temporal split is applied: training data includes loans issued before 2016, while test data consists of loans issued from 2016 onward. A clear distribution shift is observed between training and test periods, with the default rate increasing from 18.4

In particular, distribution shift affects probability calibration and may alter how participants interpret model outputs. To mitigate leakage and preserve experimental validity, all pre-processing steps (including binning and base rate estimation) are performed using training data only. However, this shift is intentionally preserved, as it reflects realistic decision-making conditions rather than a controlled i.i.d. setting.

The final feature set includes `loan_amnt`, `term`, `int_rate`, `log_annual_inc`, `dti`, `revol_util`, `home_ownership`, `purpose`, and `credit_history_years`. Feature selection prioritizes interpretability and availability at origination time over maximal predictive accuracy, consistent with the DSS objective of creating a cognitively tractable decision environment.

Two representations are maintained: continuous features are used for model training, while quantile-binned categorical representations are constructed on the training set and used for human interaction. This separation ensures that the model operates on full-resolution data while participants receive interpretable, discretized information.

Feature selection prioritizes interpretability and availability at origination time over maximal predictive accuracy, consistent with the DSS objective of creating a cognitively tractable decision environment.

For numeric features, stability is defined as a consistent (ideally monotonic) relationship with default risk. For categorical features, stability is assessed through consistent group-level differences in default rates rather than monotonicity.

Bins are constructed using training-set quantiles to ensure interpretability without introducing leakage.

Case difficulty is defined independently of model predictions using empirical base rates computed over discretized feature bins. Specifically, difficulty is derived from the distance of bin-level default rates from maximum uncertainty (0.5), ensuring that it reflects inherent ambiguity in the data rather than model output. This construction avoids circularity and allows difficulty to serve as an exogenous variable in the experimental design.

C. Experimental Design

A within-subject design is used, where each participant completes a fixed set of cases under multiple interaction protocols.

The following protocols are evaluated:

- **Human-first:** Participant makes an initial decision before seeing AI advice, then may revise.
- **AI-first:** AI prediction is shown before the participant makes a decision.
- **No-AI:** Participant makes decisions without AI assistance.

Each trial proceeds as follows:

- 1) Case features are presented
- 2) Participant provides an initial decision (if applicable)
- 3) AI prediction is revealed (protocol-dependent)
- 4) Participant provides a final decision
- 5) Confidence is reported

Cases are pre-selected using a stratified design across difficulty and model correctness, with additional constraints to ensure coverage of predicted probability ranges (confidence levels). This ensures controlled variation while avoiding combinatorial sparsity.

D. Model-Optimal Decision Gap

To evaluate the alignment between predictive models and decision-theoretic optimality, we distinguish between:

(i) the model’s implicit decision rule, which uses a fixed threshold of 0.5, and (ii) a cost-sensitive optimal decision rule defined by asymmetric decision costs.

Let τ denote the optimal threshold derived from the relative costs of false approvals and false rejections. When $\tau \neq 0.5$, model predictions and optimal decisions may systematically disagree.

We analyze how this disagreement varies across difficulty levels and construct experimental cases that expose participants to such conflicts. This creates a non-trivial decision environment in which participants must integrate probabilistic predictions with implicit or explicit decision costs.

This gap is central to the experimental design, as it allows evaluation of whether human-AI interaction improves alignment with optimal decision-making beyond the model’s default behavior.

IV. RESULTS

[This section will present findings, including model performance metrics and experimental outcomes supported by figures and tables. Results must be generated programmatically.]

V. ANALYSIS AND VALIDATION

[Interpretation of results, comparison with baselines, and discussion of project limitations. This section addresses whether the project met its research objectives.]

VI. DISCUSSION AND CONCLUSION

This study quantifies how interaction policies affect realized performance. By isolating the interaction structure, differences in decision outcomes can be attributed to how model outputs are integrated rather than the model itself. Future work will explore the non-linear relationship between model correctness and reliance across uncertainty levels.

REFERENCES

- [1] G. Bansal et al., "Does the Whole Exceed the Sum of its Parts?," in *CHI Conference on Human Factors in Computing Systems*, 2021.

APPENDIX A

FORMAL METRIC DEFINITIONS

Metric–Hypothesis Alignment. Metrics are selected to directly correspond to the hypotheses. Accuracy and reliance-based metrics directly correspond to H1–H4, while calibration-related metrics support H5–H7 and are treated as secondary analyses.

Let $y_i \in \{0, 1\}$ denote the true outcome for case i . Let $\hat{p}_i^{\text{model}}$ denote the model-predicted probability, and $\hat{p}_{i,k}^{\text{system}}$ the participant's final probability under protocol k . Let N denote the number of cases.

A. Model-Level Performance

$$\hat{y}_i^{\text{model}} = \mathbf{1} \{ \hat{p}_i^{\text{model}} > 0.5 \} \quad (1)$$

$$P_{\text{model}} = \frac{1}{N} \sum_{i=1}^N \mathbf{1} \{ \hat{y}_i^{\text{model}} = y_i \} \quad (2)$$

$$\text{Brier}_{\text{model}} = \frac{1}{N} \sum_{i=1}^N (\hat{p}_i^{\text{model}} - y_i)^2 \quad (3)$$

B. System-Level Performance

$$\hat{y}_{i,k}^{\text{system}} = \mathbf{1} \{ \hat{p}_{i,k}^{\text{system}} > 0.5 \} \quad (4)$$

$$P_{\text{system},k} = \frac{1}{N} \sum_{i=1}^N \mathbf{1} \{ \hat{y}_{i,k}^{\text{system}} = y_i \} \quad (5)$$

C. Performance Gap

$$\text{Gap}_k = P_{\text{model}} - P_{\text{system},k} \quad (6)$$

D. Performance Translation Efficiency

Let P_{human} denote performance based on initial (pre-AI) participant predictions.

$$\text{Efficiency}_k = \frac{P_{\text{system},k} - P_{\text{human}}}{P_{\text{model}} - P_{\text{human}}} \quad (7)$$

E. Weight of Advice

$$\text{WOA}_i = \frac{\hat{p}_i^{\text{final}} - \hat{p}_i^{\text{init}}}{\hat{p}_i^{\text{model}} - \hat{p}_i^{\text{init}}} \quad (8)$$

WOA is undefined when $\hat{p}_i^{\text{model}} = \hat{p}_i^{\text{init}}$; such cases are excluded.

F. Over-Reliance

$$\text{OverReliance}_i = \mathbf{1} \{ \hat{y}_i^{\text{model}} \neq y_i \} \cdot \mathbf{1} \{ \hat{y}_i^{\text{final}} = \hat{y}_i^{\text{model}} \} \quad (9)$$

G. Selective Reliance

$$M_i = \mathbf{1} \{ \hat{y}_i^{\text{model}} = y_i \} \quad (10)$$

$$\text{SelectiveReliance} = \mathbb{E} [\text{WOA}_i \mid M_i = 1] - \mathbb{E} [\text{WOA}_i \mid M_i = 0] \quad (11)$$

H. Confidence–Accuracy Correlation

$$\rho_{\text{conf,acc}} = \text{Corr}(C_i, A_i) \quad (12)$$

I. Expected Calibration Error

$$\text{ECE} = \sum_{b=1}^B \frac{|B_b|}{N} |\text{acc}(B_b) - \text{conf}(B_b)| \quad (13)$$

APPENDIX B

MIXED-EFFECTS MODELING SPECIFICATION

To evaluate the hypotheses, we estimate the following mixed-effects model:

$$\begin{aligned} \text{Correct}_{ijk} \sim & \beta_0 + \beta_1 \text{MC}_i + \beta_2 \text{Protocol}_k + \beta_3 \text{Diff}_i \\ & + \beta_4 (\text{MC}_i \times \text{Protocol}_k) \\ & + \beta_5 (\text{MC}_i \times \text{Diff}_i) \\ & + \beta_6 (\text{Protocol}_k \times \text{Diff}_i) \\ & + \beta_7 (\text{MC}_i \times \text{Protocol}_k \times \text{Diff}_i) \\ & + u_j + v_i \end{aligned} \quad (14)$$

where MC_i denotes model correctness, Protocol_k the interaction protocol, and Diff_i the difficulty level.

This specification allows testing whether representation and difficulty moderate the effect of model correctness on final decision accuracy.

APPENDIX C

NORMATIVE DECISION MODEL

We define a normative benchmark to evaluate whether human–AI decisions align with optimal decision-making under uncertainty.

Let $\hat{p}_i^{\text{model}} = P(y_i = 1 \mid x_i)$ denote the predicted probability of default.

A rational decision-maker should approve a loan if:

$$\hat{p}_i^{\text{model}} < \tau \quad (15)$$

where τ is a decision threshold determined by relative costs of false approvals and false rejections.

More generally, let $U(a, y)$ denote the utility of action $a \in \{\text{approve, reject}\}$ under outcome y . The optimal decision rule is:

$$a_i^* = \arg \max_a \mathbb{E}[U(a, y_i) \mid x_i] \quad (16)$$

This framework allows us to evaluate whether participants:

- follow optimal decision rules
- deviate due to misinterpretation of probabilities

- exhibit systematic biases in integrating AI advice

Deviation from the normative benchmark provides a principled measure of decision quality beyond raw accuracy.

We operationalize the decision threshold as:

$$\tau = \frac{C_{\text{FP}}}{C_{\text{FP}} + C_{\text{FN}}} \quad (17)$$

where C_{FN} is the cost of approving a defaulting loan and C_{FP} is the cost of rejecting a non-defaulting loan. This yields a cost-sensitive decision rule that may differ substantially from the standard 0.5 threshold.

APPENDIX D LOGGING SCHEMA

Each trial logs the following variables:

Identifiers: participant ID, trial index, case ID

Experimental Conditions: protocol (AI-first, human-first, no-AI), difficulty tier

Model Outputs: true label y_i , predicted probability $\hat{p}_i^{\text{model}}$, model decision, model correctness

Human Decisions: initial decision (when applicable), final decision, confidence rating

Behavioral Measures: decision change indicator, weight-of-advice (WOA), agreement with model

Derived Metrics: over-reliance, beneficial override, deviation from optimal decision

The system records detailed behavioral logs for each trial, including initial and final decisions, confidence ratings, decision revisions, and decision time. These logs enable analysis of reliance behavior, calibration, and interaction effects.